

Question	Answer	Marks
4(a)	(two) waves (travelling at same speed) in opposite directions overlap	B1
	(waves are same type and) have same frequency / wavelength	B1
4(b)(i)	$v = f\lambda$ $f = 330 / 0.18$	C1
	= 1800 Hz (1830 Hz)	A1
4(b)(ii)	$T = 1 / 1800 (= 5.5 \times 10^{-4})$ time-base setting = $(1.5 \times 5.5 \times 10^{-4}) / 8.0$ or $1 / (1800 \times 5.3)$	C1
	= $1.0 \times 10^{-4} \text{ s cm}^{-1}$	A1
4(b)(iii)	waveform drawn with same period as original waveform	B1
	waveform drawn with amplitude of 1.7 cm	B1
4(c)(i)	distance = $\lambda / 2 = 0.18 / 2$ = 0.090 m	A1

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4(c)(ii)	letter N shown at level B and at level A and not anywhere else.	B1
4(c)(iii)	$m = \rho Ax$ $= 0.79 \times 13 \times 9.0 (=92.4)$ or $790 \times 13 \times 10^{-4} \times 0.090 (=0.0924)$ $t = 92.4 / 6.7$ or $0.0924 / 0.0067$	C1
	$= 14 \text{ s}$	A1

Question	Answer	Marks
5(a)	sum of e m f (s) = sum of p d (s)	M1